

Sudden Hearing Loss

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Sudden hearing loss is moderate to severe hearing loss that develops over about 72 hours or is noticed on awakening.

Such hearing loss typically affects only one ear (unless the cause is a medication). Depending on the cause of sudden hearing loss, people may have other symptoms such as ringing in the ears ([tinnitus](#)), dizziness, or a false sensation of spinning or moving ([vertigo](#)). About 1 in 5,000 to 1 in 10,000 people each year develop sudden hearing loss. For hearing loss that develops gradually, see [Hearing Loss](#).

Causes of Sudden Hearing Loss

Causes of sudden hearing loss fall into 3 general categories:

- An unknown cause
- An obvious explanatory event (such as a brain infection or head injury)
- An underlying disorder

Unknown cause

In most people, no cause can be found for their sudden hearing loss. However, doctors have several theories. Possible causes include [viral infections](#) (particularly infections with [herpes simplex virus](#)), an attack on the inner ear or its nerves by the body's immune system ([autoimmune reaction](#)), and blockage of the small blood vessels of the inner ear or the blood vessels of its nerves. Perhaps different causes affect different people.

Obvious event

In many other people, a cause for the sudden hearing loss is obvious. Such causes include

- [Head injury](#)
- [Severe pressure change](#)
- Medications that damage the inner ear ([ototoxic medications](#))
- Infections

Head injury (such as a fracture of the temporal bone in the skull or sometimes a severe concussion without a fracture) can damage the inner ear and cause sudden hearing loss.

Severe pressure changes (such as those that can occur with diving or less often by bearing down during weightlifting) can cause a hole (fistula) to form between the middle and inner ear. Sometimes, such a fistula is present from birth and can spontaneously cause sudden hearing loss or make the person more susceptible to hearing loss after a head injury or pressure changes.

Ototoxic medications are medications that have damaging side effects to the ears. Some medications can rapidly cause hearing loss, sometimes within a day (especially if the dose given is too high). A few people have a rare genetic disorder that makes them more susceptible to hearing loss caused by the class of antibiotics called aminoglycosides. All people taking such antibiotics should have blood tests while taking the medication to screen for toxic levels that can cause hearing loss.

A number of **infections** cause sudden hearing loss during or immediately after acute illness. These infections include [bacterial meningitis](#), [Lyme disease](#), and many viral infections. The most common viral causes in the developed world are [mumps](#) and [herpes simplex](#) brain infection. [Measles](#) is a very rare cause because most people are immunized against the infection.

Underlying disorders

Sudden hearing loss rarely can be the first symptom of some disorders that usually have other initial symptoms. Such disorders include a tumor of the auditory nerve called [vestibular schwannoma](#), [multiple sclerosis](#), [Meniere disease](#), or a small [stroke](#) due to a blockage of a branch of the artery that goes to the balance center of the brain (the cerebellum). Sometimes a [syphilis infection](#) reactivates in people who have [HIV infection](#). This reactivation can cause sudden hearing loss.

Rarer disorders include [Cogan syndrome](#), in which an autoimmune reaction attacks the inner ear (and also the surface of the eye); other disorders also involving blood vessel inflammation ([vasculitis](#)); and blood disorders such as [macroglobulinemia](#), [sickle cell disease](#), and some forms of [leukemia](#).

Evaluation of Sudden Hearing Loss

The following information can help people decide when a doctor's evaluation is needed and help them know what to expect during the evaluation.

Warning signs

Sudden hearing loss is itself a warning sign.

When to see a doctor

Anyone with sudden hearing loss should see a doctor right away because some causes must be treated quickly. If symptoms of dysfunction of the nervous system other than hearing loss are present, hearing loss may be a symptom of nerve or brain dysfunction.

What the doctor does

Doctors first ask questions about the person's symptoms and medical history. Doctors then do a physical examination. What they find during the history and physical examination may suggest a cause of the sudden hearing loss and the tests that may need to be done.

Doctors note whether hearing loss affects one or both ears and whether a specific event such as [head injury](#), [diving injury](#), or infectious illness occurred. They ask about accompanying symptoms that involve the ear (such as ringing in the ears or ear discharge), balance center (such as disorientation in the dark or vertigo), and other parts of the brain and nervous system (such as headache, weakness, or an abnormal sense of taste). They try to identify whether people are currently taking (or recently took) any [ototoxic medications](#).

The physical examination focuses on the ears and hearing and on examination of the nervous system.



Some Causes and Features of Sudden Hearing Loss

Cause	Common Features*	Diagnostic Approach†
Unknown		
—	Deafness in only one ear No other symptoms	MRI using a contrast agent (gadolinium)
Obvious causes		
	Deafness in one or both ears In people with a serious, acute illness	
Acute infection (such as bacterial meningitis , Lyme disease , mumps , or herpes simplex)	Often headache and confusion With Lyme disease, deafness preceded by a typical rash and flu-like symptoms With mumps, pain in cheeks with swallowing	If not already done, blood tests and a spinal tap (lumbar puncture)
Head injury	Deafness usually in only one ear Sometimes fluid (bloody, blood-tinged, or clear) coming from the affected ear	CT and/or MRI
Pressure changes (as may occur during diving)	Deafness in one or both ears Sudden onset during causative activity (for example, scuba diving, rapid descent in	Tympanometry (placement of a device in the ear to measure how well sound passes through the ear)

